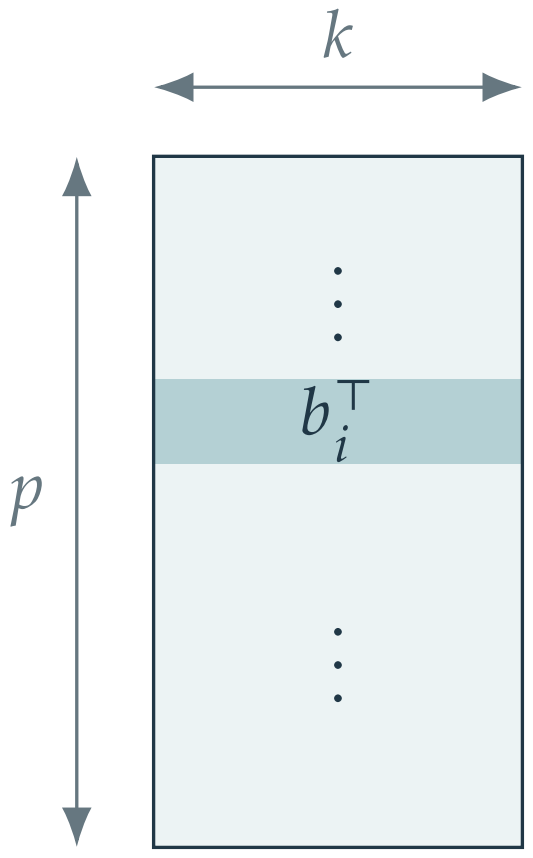

$$B = V^\top S, \quad B^\top B = I_k$$

$$\beta_i = \|b_i\|^2 \geq 0, \quad \sum_i \beta_i = k$$